

CSE470: Software Engineering

Layered Architecture



Monolithic Software

- The end product come at end of the process model
- There is no separation of concern (code) of different software components.
- All code may be written in a single file with html, sql queries, logic checking etc.



Problems of Monolithic Software

- No separation between components
- Changing a component affects other components.

For example – What if I want to change the UI from JavaScript to Angular ?



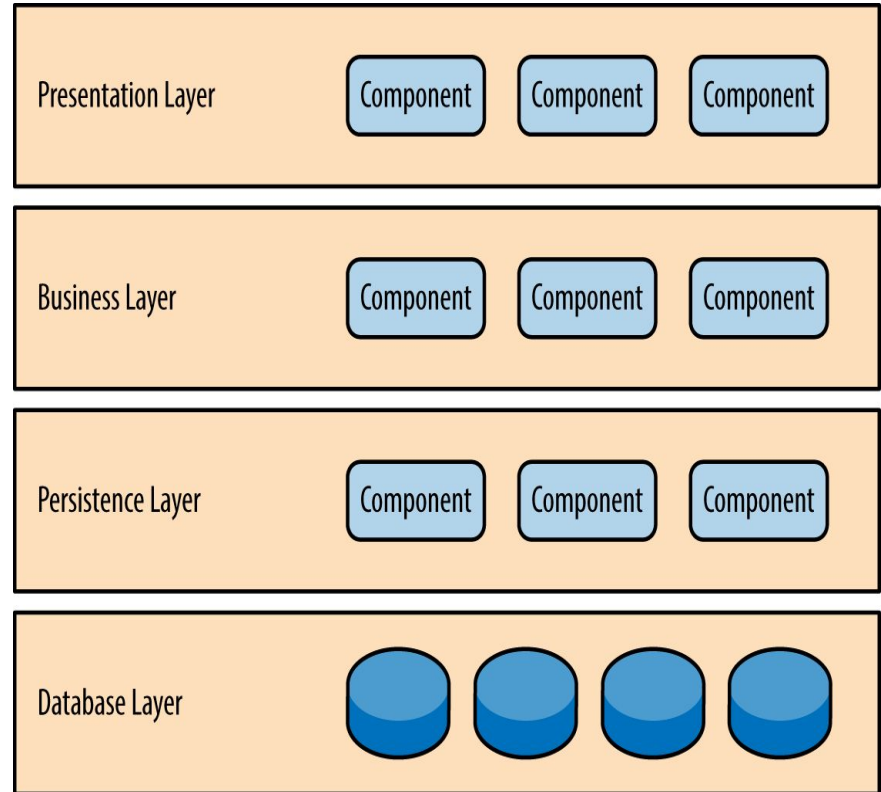
What is Layered Architecture

- Organises the system into a set of layers (or abstract machines) each of which provide a set of services.
- Supports the incremental development of sub-systems in different layers. When a layer interface changes, **only the adjacent layer is affected.**



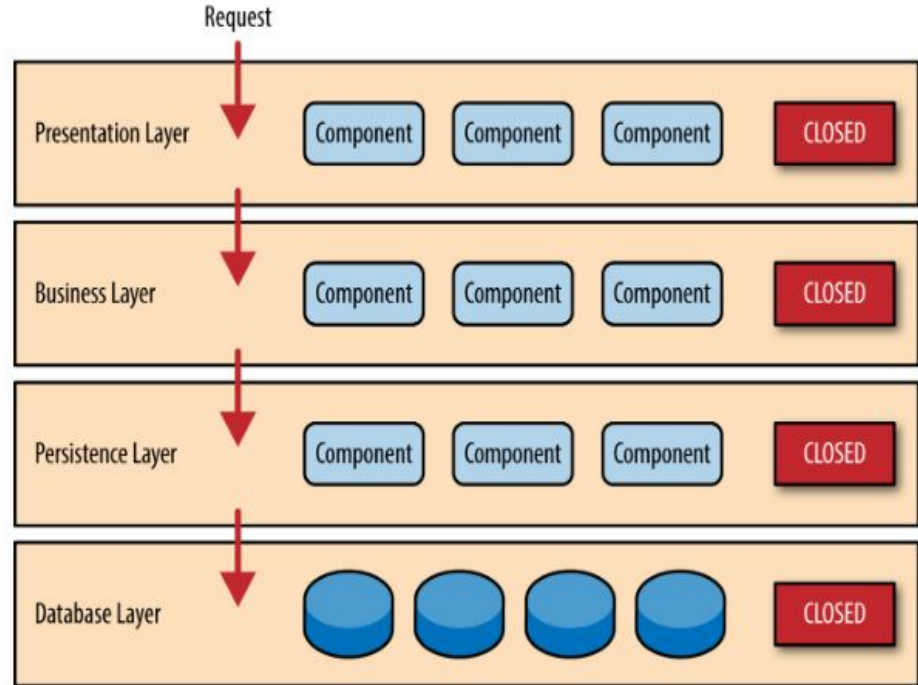
Layered Architecture

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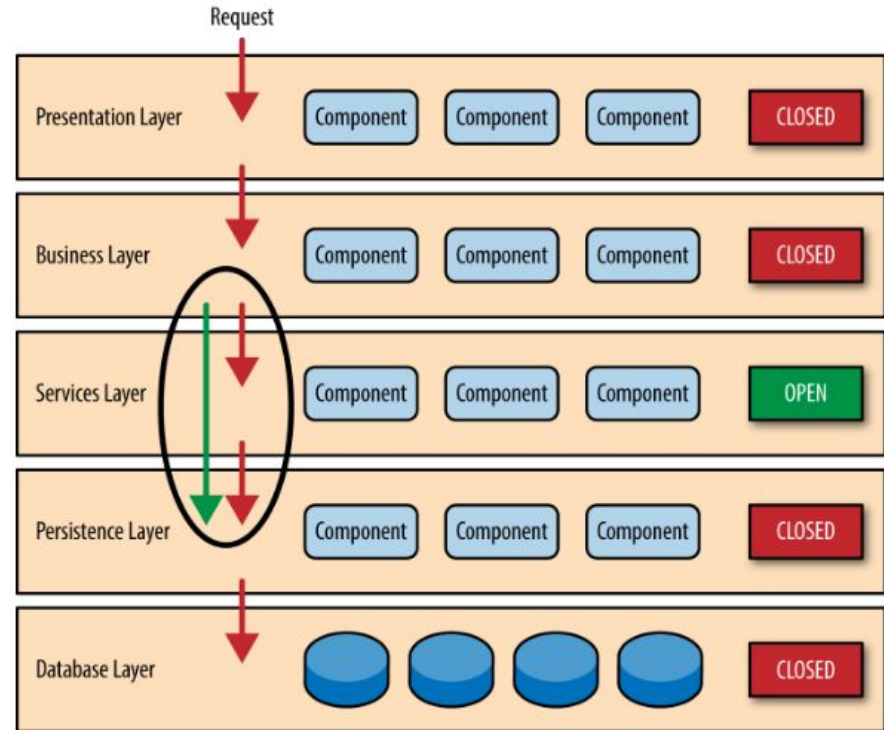
Layered Architecture

- Layers can be open or **closed**
- A closed layer can only be accessed by the layer above.
- A change in one layer does not affect others. It provides isolation.
- However, what if we want to add a new layer where shared utilities will be provided to be used by the Business Layer. But, we need to use it sometimes.

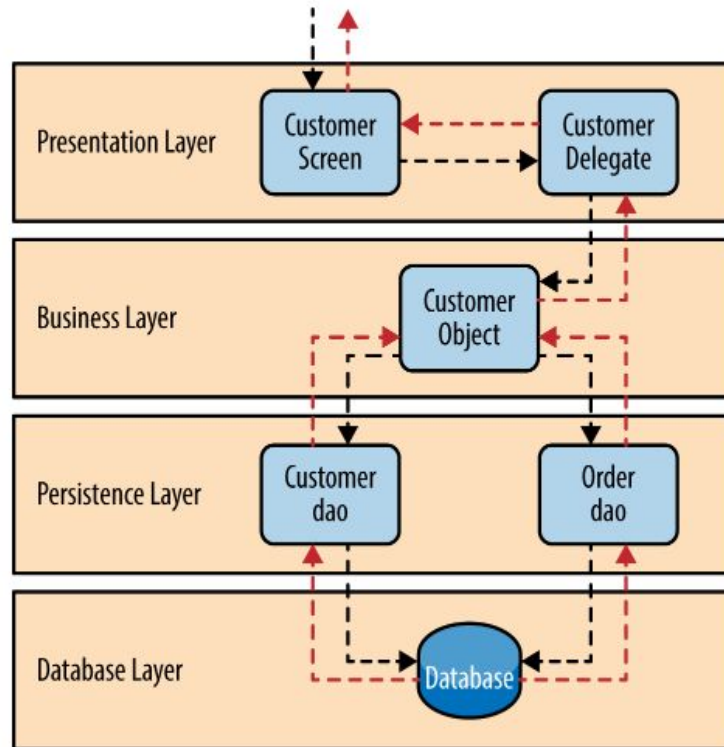


Layered Architecture

- Here comes the concept of open layers.
- An open can be bypassed by upper layers.
- Too many open layers may affect the actual essence of layered architecture.



Example



dao = data access object

When Used

- Used when building new facilities on top of existing systems
- When the development is spread across several teams with each team responsible for a layer of functionality
- When there is a requirement for multi-level security (for instance, any secured transaction based system)



Advantages-Disadvantages

Advantages

- Allows easy replacement or addition of entire layers so long as the interface is maintained.
- Testing is easy as components are isolated
- Parallel team work

Disadvantages

- In practice, providing a clean separation between layers is often difficult and a high-level layer may have to interact directly with lower-level layers rather than through the layer immediately below it.
- Performance can be a problem because of multiple levels of interpretation of a service request as it is processed at each layer.
- Parallel processing is not possible (Sequential structure)
- Example: OSI model (7 layers)

